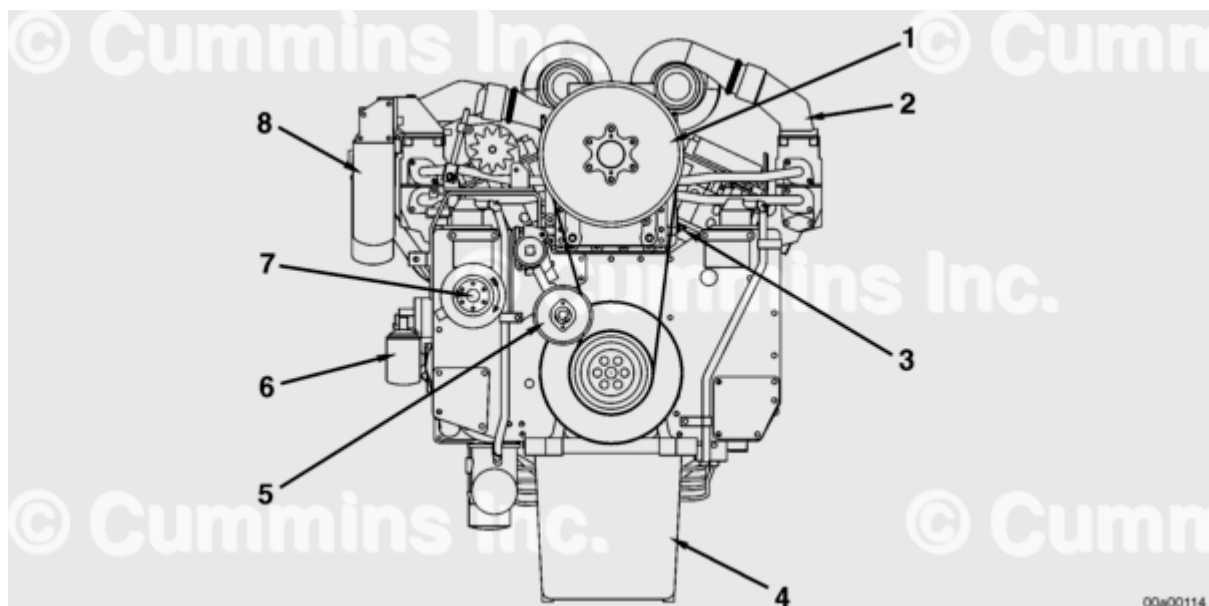


Trainings ▾

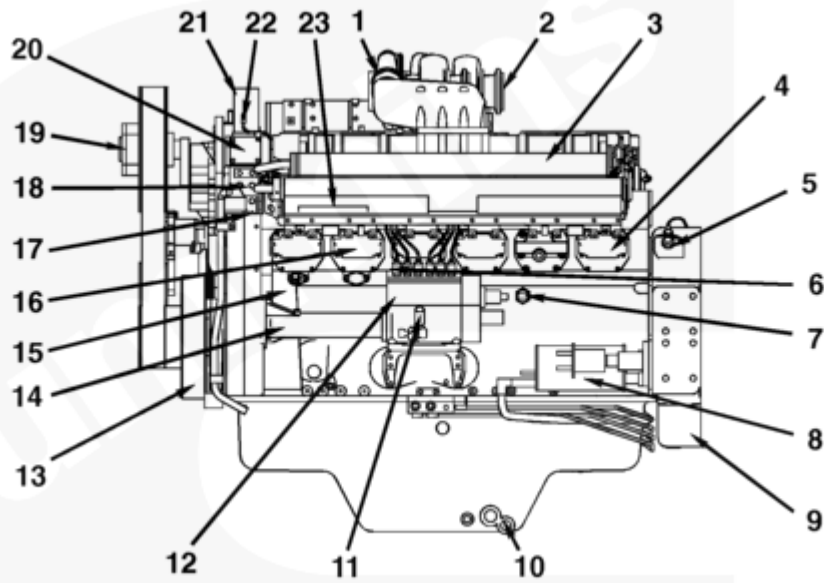
Engine Views

The following illustrations contain information about engine components, filter locations, drain points, and access locations for instrumentation and engine controls. The information and configuration of components shown in these drawings are of a general nature. Some component locations will vary depending on applications and installations.



Front View - Aftercooled

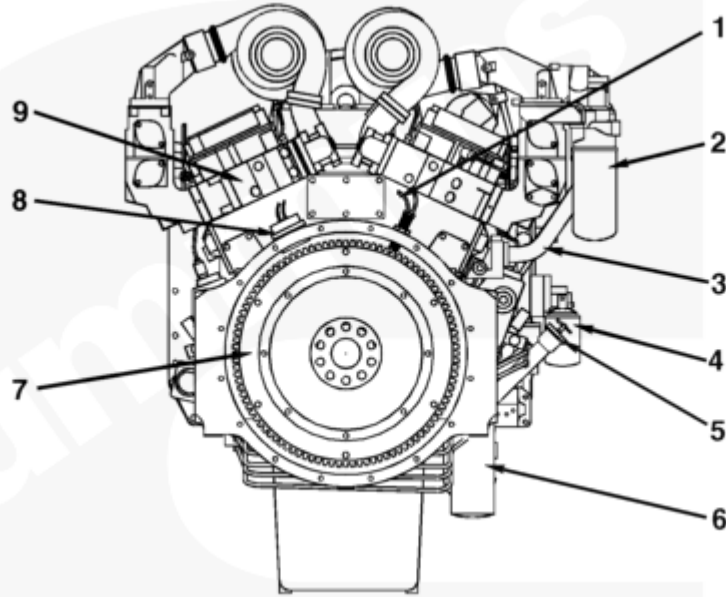
1. Fan hub
2. Air crossover
3. Coolant temperature sensor
4. Oil pan
5. Fan idler tensioner pulley
6. Water filters
7. Accessory drive
8. Full-flow or combination oil filters.



00a00111

Left View - Aftercooled

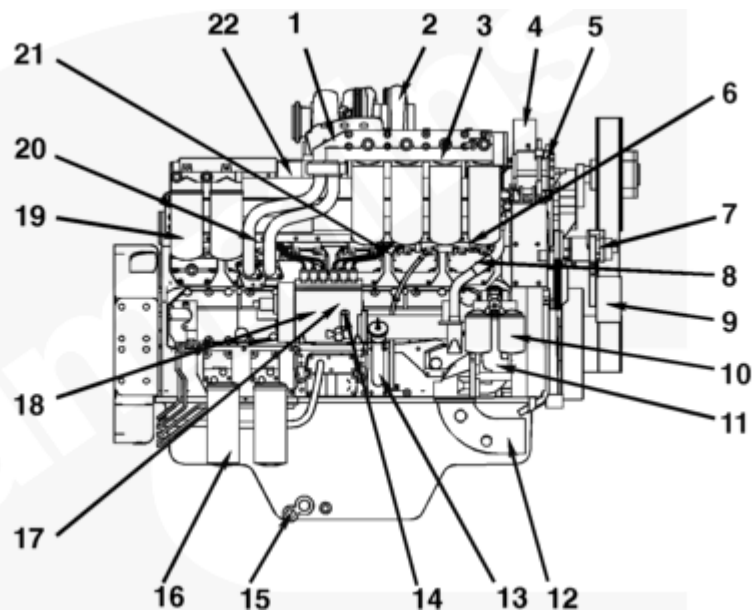
1. Turbocharger inlet connection
2. Turbocharger outlet connection
3. Aftercooler housing (intake manifold on charge air cooled engines)
4. Cam follower cover
5. Engine speed sensor (engine position sensor industrial engines)
6. High-pressure fuel supply lines
7. Oil pressure sensor
8. Prelubricating starter (industrial engines **only**)
9. Flywheel housing
10. Lubricating oil drain
11. Fuel lift pump
12. Fuel injection pump overflow valve
13. Vibration damper
14. Fuel pump drive
15. ECM (industrial **only**)
16. Industrial data link (J1587 and 2 6-pin Deutsch™ connectors, or J1939/J11587 left bank and 1 9-pin Deutsch™ connector)
17. Coolant temperature sensor
18. Crankcase breather
19. Fan hub
20. Thermostat housing
21. Water outlet connection
22. Water vent tubes
23. Engine dataplate.



00a00113

Rear View - Aftercooled

1. Engine speed sensor (industrial)
2. Lubricating oil bypass filters (when equipped with full-flow oil filters)
3. Lubricating oil transfer tube
4. Water filters
5. Lubricating oil filler tube
6. Fuel filters
7. Flywheel
8. Engine position sensor (industrial)/engine speed sensor (generator-drive, generator set)
9. Cylinder head.

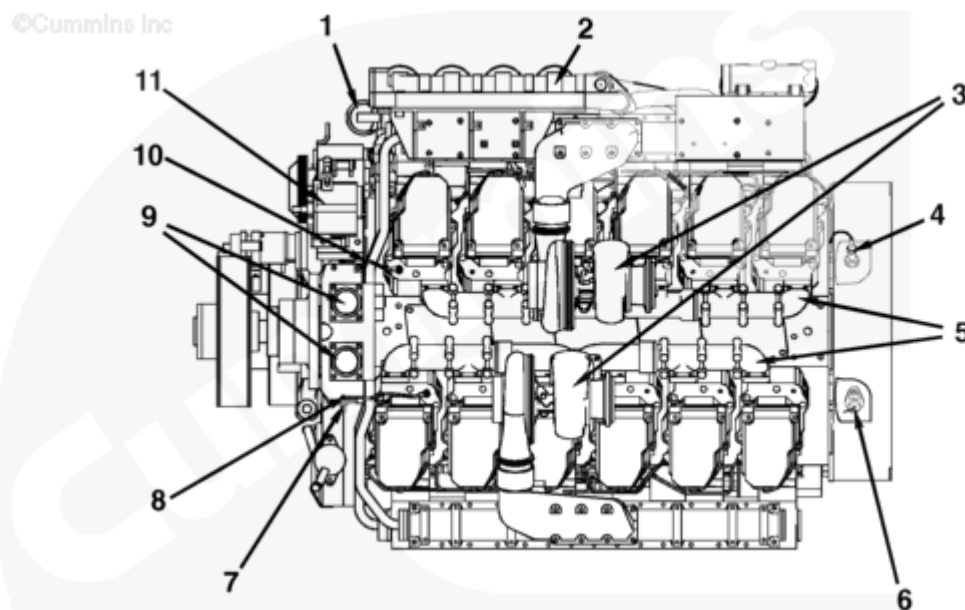


00a00110

Right View - Aftercooled

1. Air crossover housing
2. Turbocharger

3. Full-flow or combination oil filters
4. Water outlet connection
5. Lifting bracket
6. ECM (industrial **only**)
7. Fan belt idler assembly
8. Aftercooler water inlet tube
9. Fan belt idler pulley
10. Water filter
11. Water pump
12. Water inlet connection
13. Lubricating oil filler tubes
14. Fuel lift pump
15. Lubricating oil drain
16. Fuel filters
17. Fuel injection pump overflow valve
18. Fuel injection pump
19. Lubricating oil bypass filters (when equipped with full-flow oil filters)
20. Lubricating oil transfer tube
21. J1939 3-pin data link (industrial engines)
22. Air intake manifold.

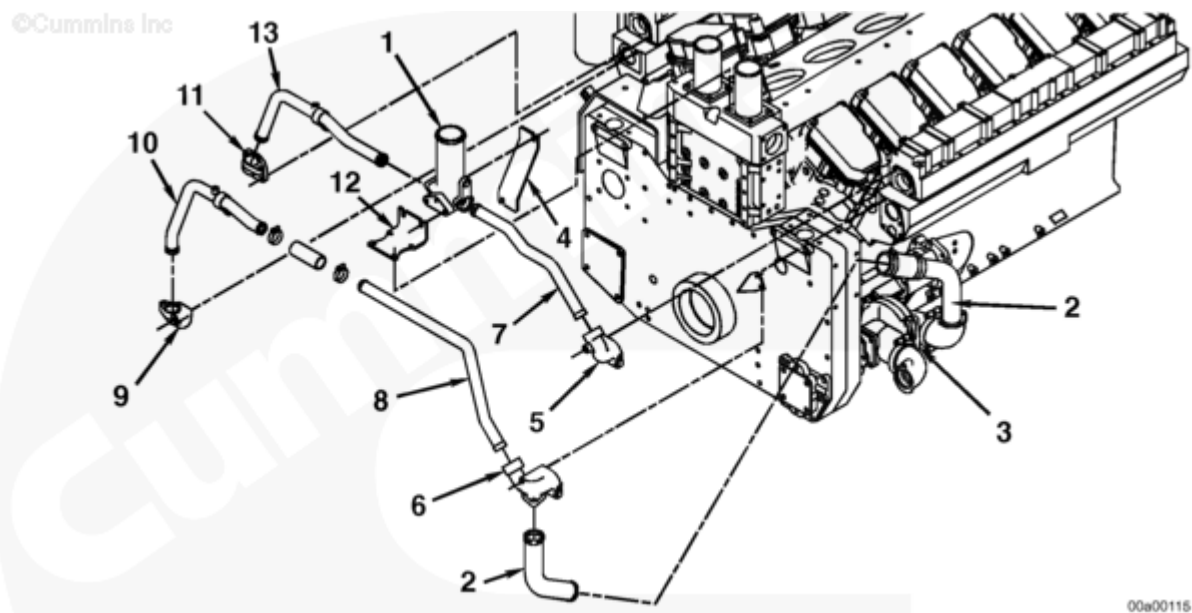


Top View - Aftercooled

00a00112

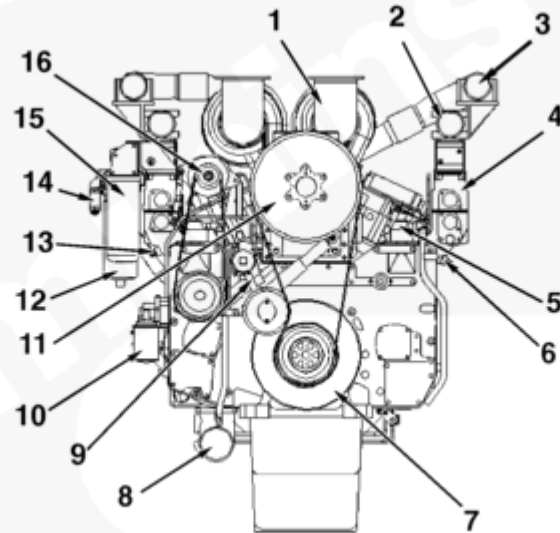
1. Water filter
2. Full-flow or combination oil filters
3. Turbochargers
4. Engine speed sensor (industrial)
5. Exhaust manifolds
6. Engine position sensor (industrial)/engine speed sensor (generator-drive, generator set)
7. Coolant temperature sensor
8. Water vent connection

- 9. Water outlet connection
- 10. Water vent connection
- 11. Alternator.



Low-Temperature Aftercooled - Two Pump Two Loop Cooling System

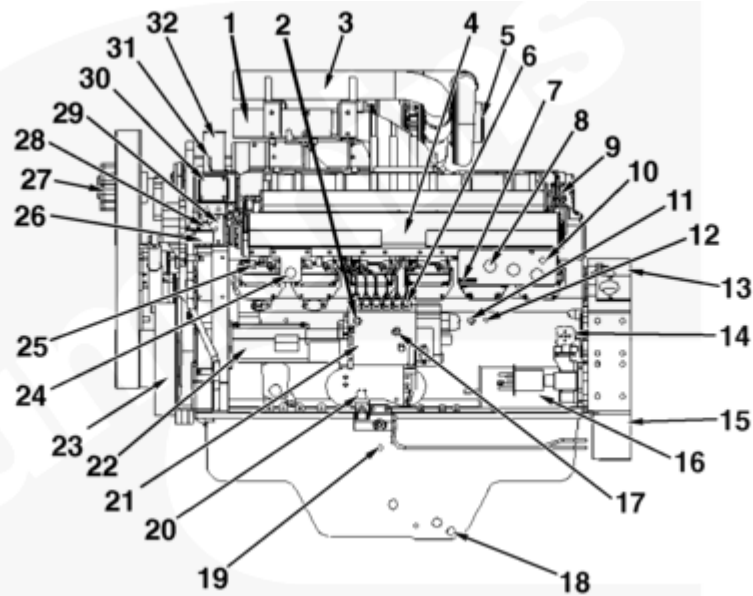
- 1. Water manifold
- 2. Water transfer tube
- 3. Water pump and spline coupling sleeve
- 4. Manifold brace
- 5. Water transfer connection
- 6. Water transfer connection
- 7. Water transfer tube
- 8. Water transfer tube
- 9. Water transfer connection
- 10. Water transfer tube and tube clamp
- 11. Water transfer connection
- 12. Manifold bracket
- 13. Water transfer tube.



00a00118

Front View - Charge Air Cooled Generator Control System

1. Turbocharger exhaust connection
2. Piping from charge air cooler
3. To charge air cooler
4. Aftercooler housing/intake manifold
5. Crankcase breather
6. Intake manifold pressure sensor 1 (left bank)
7. Vibration damper
8. Coolant inlet connection
9. Fan belt idler assembly
10. Coolant additive filter
11. Fan hub
12. Fuel filter
13. Intake manifold pressure sensor 2 (right bank)
14. Fuel inlet fitting
15. Combination lubricating oil filters
16. Alternator.



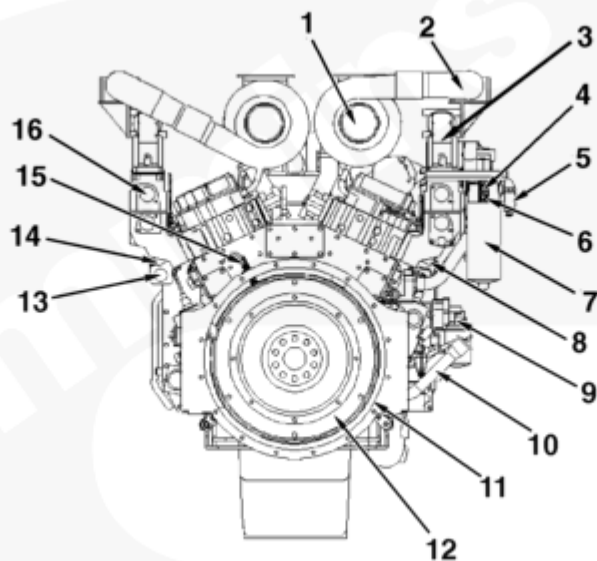
00a00117

Left View - Charge Air Cooled Generator Control System

1. From charge air cooler
2. Fuel injection pump overflow valve
3. To charge air cooler
4. Aftercooler housing/intake manifold
5. Turbocharger inlet connection
6. High-pressure fuel supply lines
7. Barometric/ambient pressure sensor
8. INLINE™ connectors
9. Heater connector
10. Intake manifold temperature sensor 1 (left bank)
11. Oil pressure sensor
12. Oil temperature sensor
13. Engine speed sensor
14. Starter solenoid
15. Flywheel housing
16. Starter
17. Overflow valve
18. Oil drain port
19. Oil level sensor (optional)
20. Fuel shutoff valve 1 (left bank)
21. Left bank fuel pump
22. Fuel pump drive housing
23. Vibration damper
24. Data link connector (9-pin Deutsch™ connector)
25. Intake manifold pressure sensor 1 (left bank)
26. Crankcase breather
27. Fan hub
28. Coolant pressure sensor
29. Coolant temperature sensor

- 30. Thermostat housing
- 31. Coolant vent tube
- 32. Coolant outlet connection.

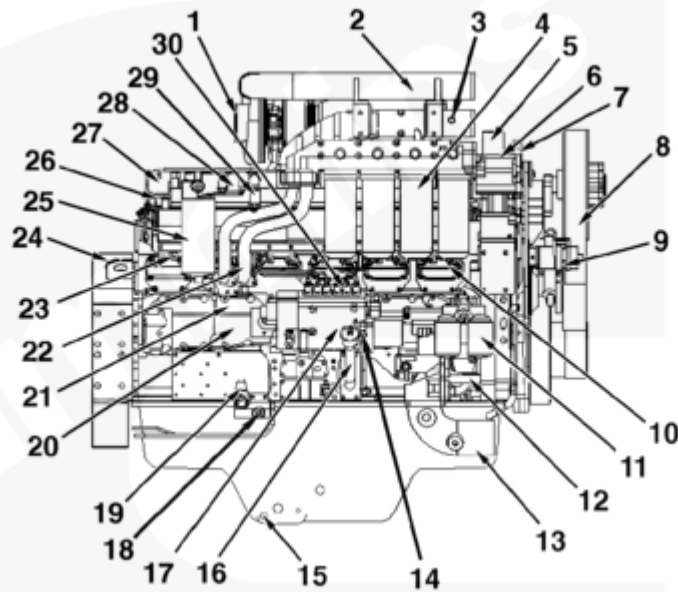
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00a00119

Rear View - Charge Air Cooled Generator Control System

- 1. Turbocharger inlet connection
- 2. Piping to charge air cooler
- 3. Piping from charge air cooler
- 4. Fuel supply pressure sensor
- 5. Fuel inlet fitting
- 6. Fuel supply temperature sensor
- 7. Fuel filter
- 8. Intake manifold temperature sensor 2 (right bank)
- 9. Coolant filter
- 10. Oil filler tube
- 11. Flywheel housing
- 12. Flywheel
- 13. INLINE™-E connector
- 14. Intake manifold temperature sensor 1 (left bank)
- 15. Engine speed sensor
- 16. Aftercooler housing/intake manifold.



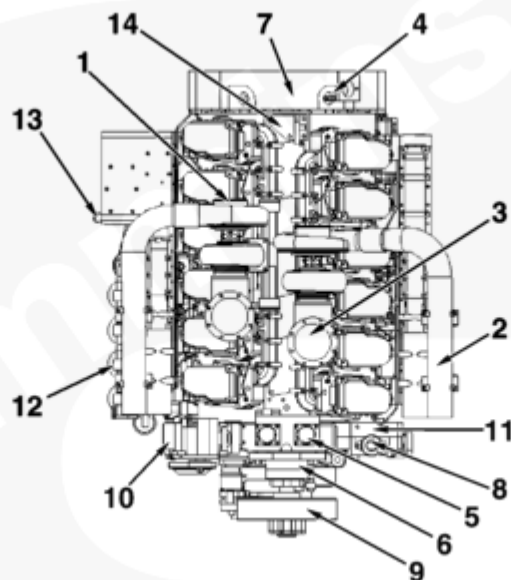
00a00116

Right View - Charge Air Cooled Generator Control System

1. Turbocharger inlet connection
2. Piping to charge air cooler
3. From the charge air cooler
4. Combination lubricating oil filters
5. Coolant outlet connection
6. Alternator
7. Lifting bracket
8. Fan hub
9. Fan belt idler assembly
10. Intake manifold pressure sensor 2 (right bank)
11. Coolant additive
12. Coolant pump
13. Coolant inlet connection
14. Fuel injection pump overflow valve
15. Oil drain port
16. Oil filler tube
17. Right bank fuel pump
18. Fuel outlet fitting
19. Fuel shutoff valve 2 (right bank)
20. Centinel™ oil tank level switch connector
21. Centinel™ driver connector
22. Oil transfer tubes
23. Intake manifold temperature sensor 2 (right bank)
24. Flywheel housing
25. Fuel filter
26. Fuel supply temperature sensor
27. Fuel supply pressure sensor
28. Fuel supply pump housing
29. Fuel inlet fitting
30. Fuel inlet fitting

30. High-pressure fuel lines.

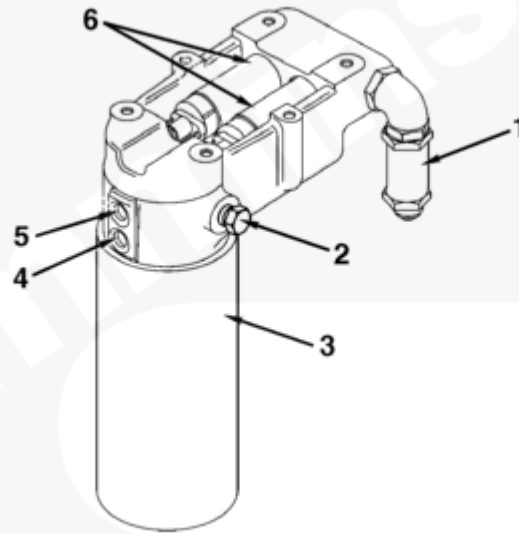
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19803652

Top View - Charge Air Cooled Generator Control System

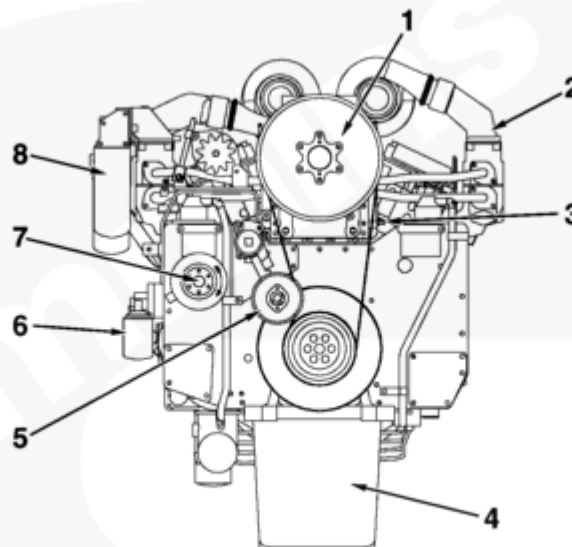
1. Turbocharger inlet connection
2. Piping to charge air cooler
3. Exhaust outlet connection
4. Engine speed sensor
5. Coolant outlet connection
6. Front lifting bracket
7. Flywheel housing
8. Crankcase breather
9. Fan hub
10. Alternator
11. Coolant level switch connector
12. Combination lube oil filters
13. Fuel inlet fitting
14. Rear lifting bracket.



19803660

Fuel Supply Pump and Filter with Electric Lift Pumps

1. Fuel supply connection
2. Fuel outlet connection
3. Fuel filter
4. Fuel supply temperature sensor
5. Fuel supply pressure sensor
6. Fuel supply pump 1 and 2.



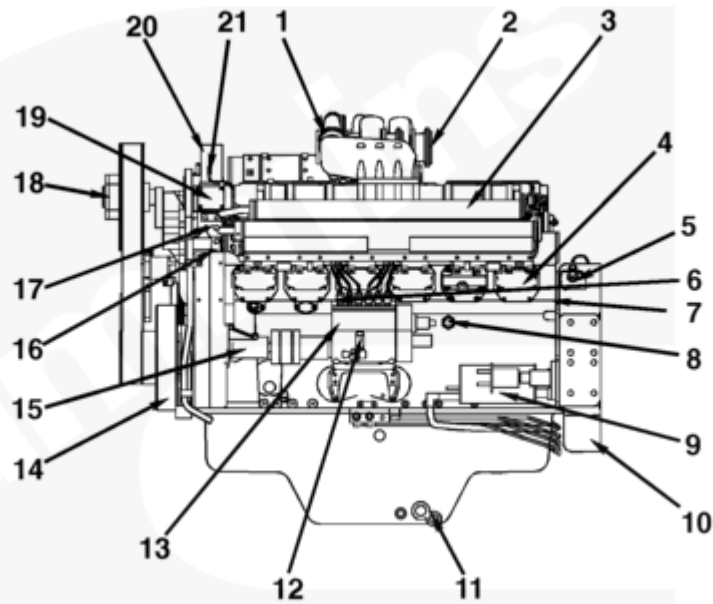
19803658

Front View - Aftercooled-QST30 Upgrade

1. Fan hub
2. Air crossover
3. Coolant temperature sensor
4. Oil pan
5. Fan idler tensioner

- 6. Water filters
- 7. Engine barring location
- 8. Full-flow oil filters.

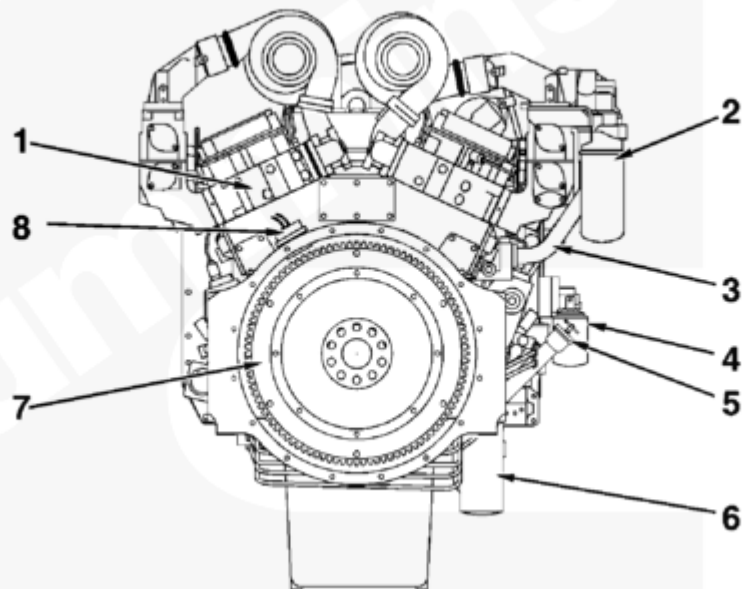
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00a00121

Left View - Aftercooled QST30 Upgrade

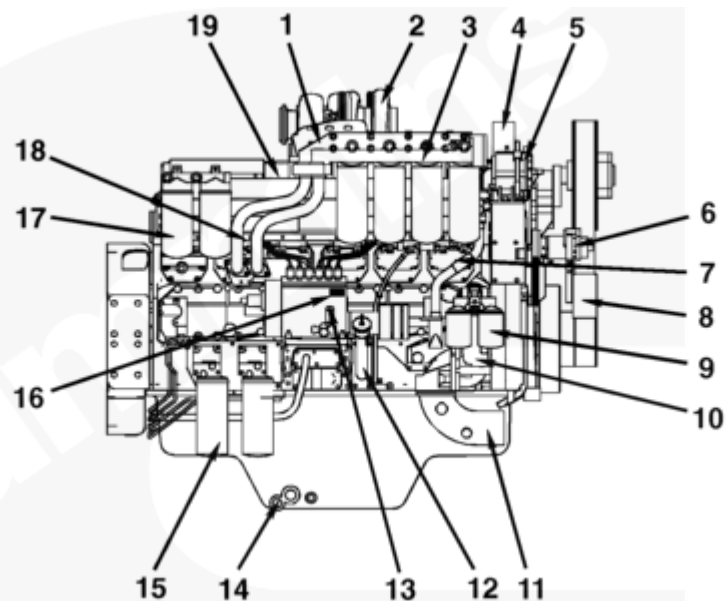
- 1. Turbocharger inlet connection
- 2. Turbocharger outlet connection
- 3. Aftercooler housing (intake manifold on charge air cooled engines)
- 4. Cam follower cover
- 5. Engine speed sensor
- 6. High-pressure fuel supply lines
- 7. Data link connector (9-pin Deutsch™)
- 8. Oil pressure sensor
- 9. Starter
- 10. Flywheel housing
- 11. Oil drain
- 12. Fuel lift pump
- 13. Fuel injection overflow valve
- 14. Vibration damper
- 15. Fuel pump drive
- 16. Coolant temperature sensor
- 17. Crankcase breather
- 18. Fan hub
- 19. Thermostat housing
- 20. Water outlet connection
- 21. Water vent tubes.



19803736

Rear View - Aftercooled QST30 Upgrade

1. Cylinder head
2. Bypass filter
3. Oil transfer tube
4. Water filters
5. Oil fill
6. Fuel filters
7. Flywheel
8. Engine speed sensor.

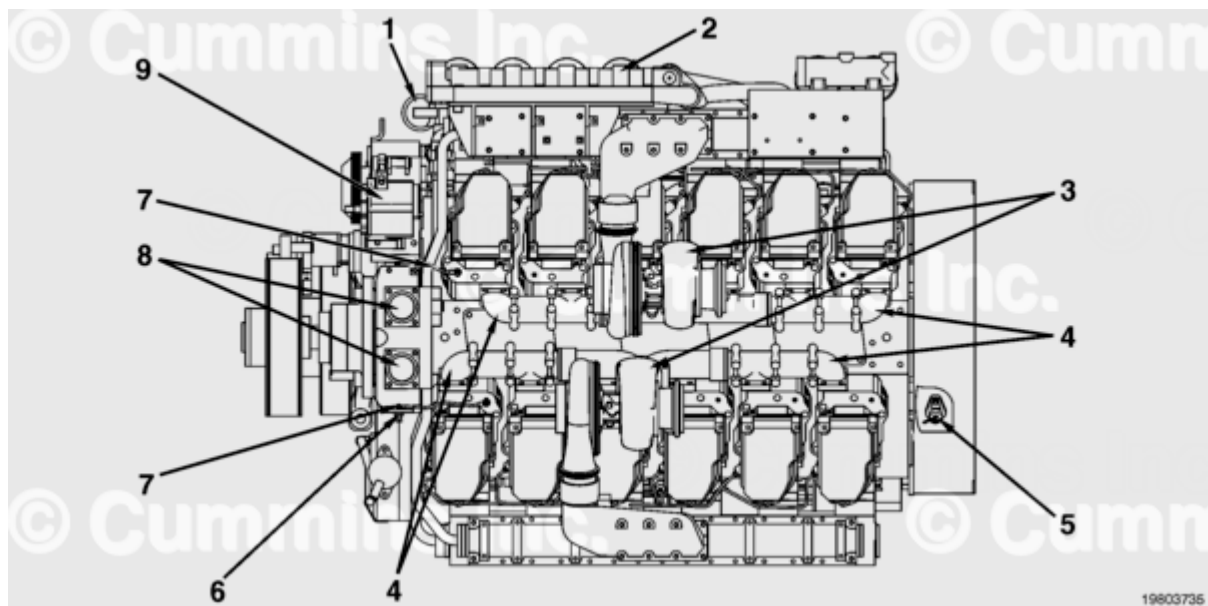


00a00120

Right View - Aftercooled QST30 Upgrade

1. Air crossover housing

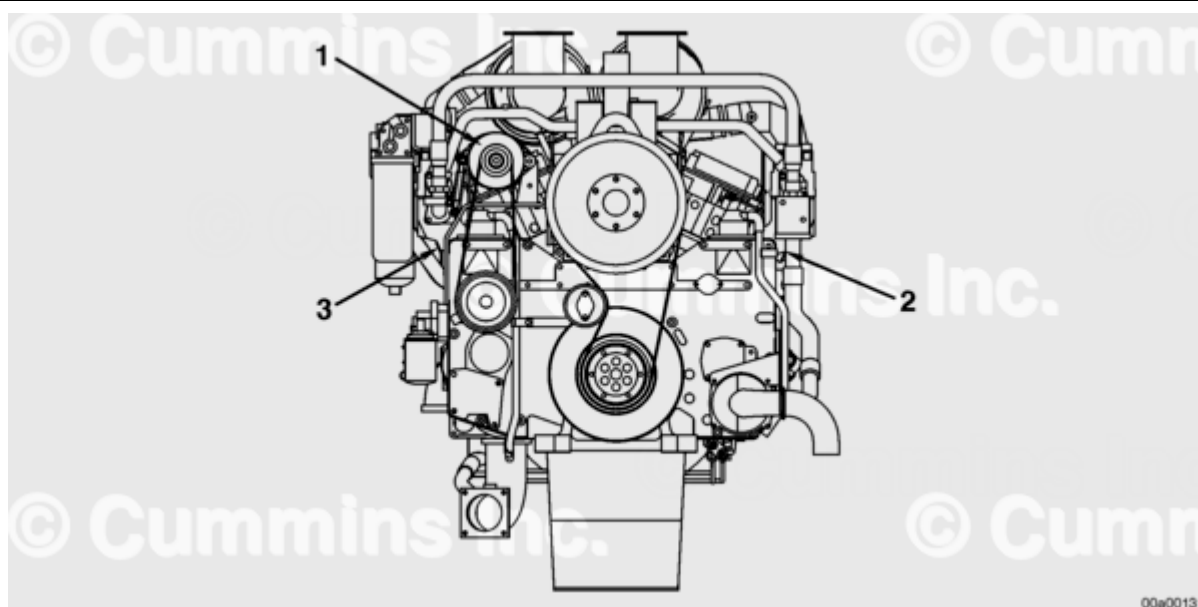
2. Turbocharger
3. Full flow oil filters
4. Water outlet connection
5. Lifting bracket
6. Fan belt idler assembly
7. Water bypass tube
8. Fan belt idler pulley
9. Water filter
10. Water pump
11. Water inlet connection
12. Oil filler tube
13. Lift pump
14. Oil drain
15. Fuel filter
16. Fuel injection pump overflow valve
17. Oil bypass filters
18. Oil transfer tube
19. Intake manifold.



Top View - Aftercooled QST30 Upgrade

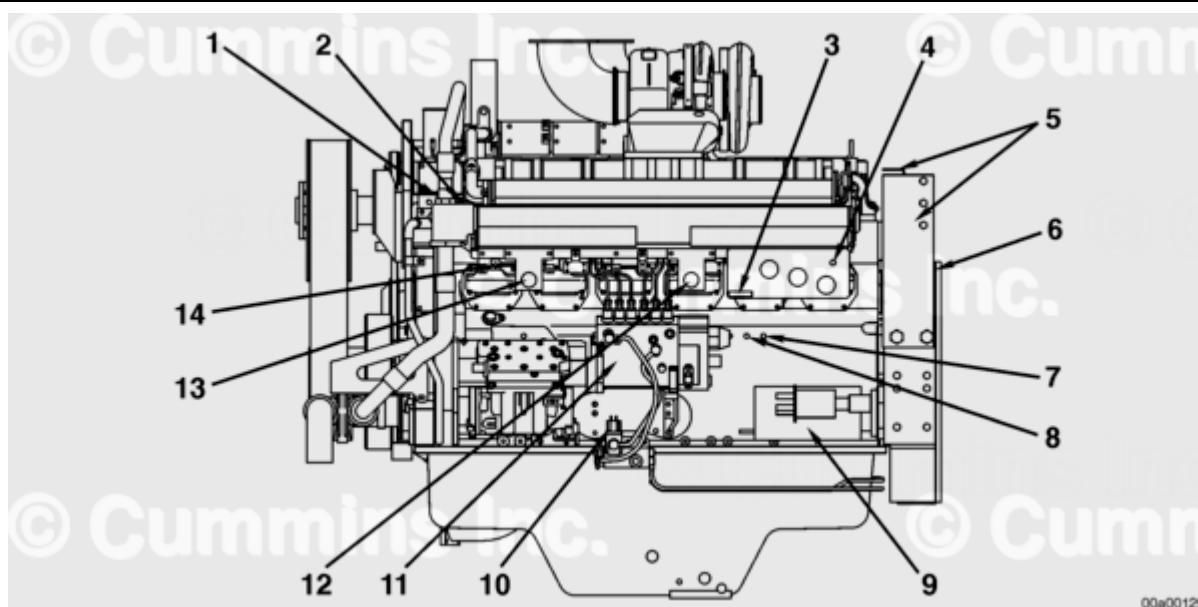
1. Water filter
2. Full-flow oil filters
3. Turbochargers
4. Exhaust manifolds
5. Engine speed sensor
6. Coolant temperature sensor
7. Water vent connection
8. Water outlet connection
9. Alternator.

The illustrations below show the locations of electronic components unique to the QST30 Power Generation Interface (PGI) Engine.



Front View - QST30 PGI

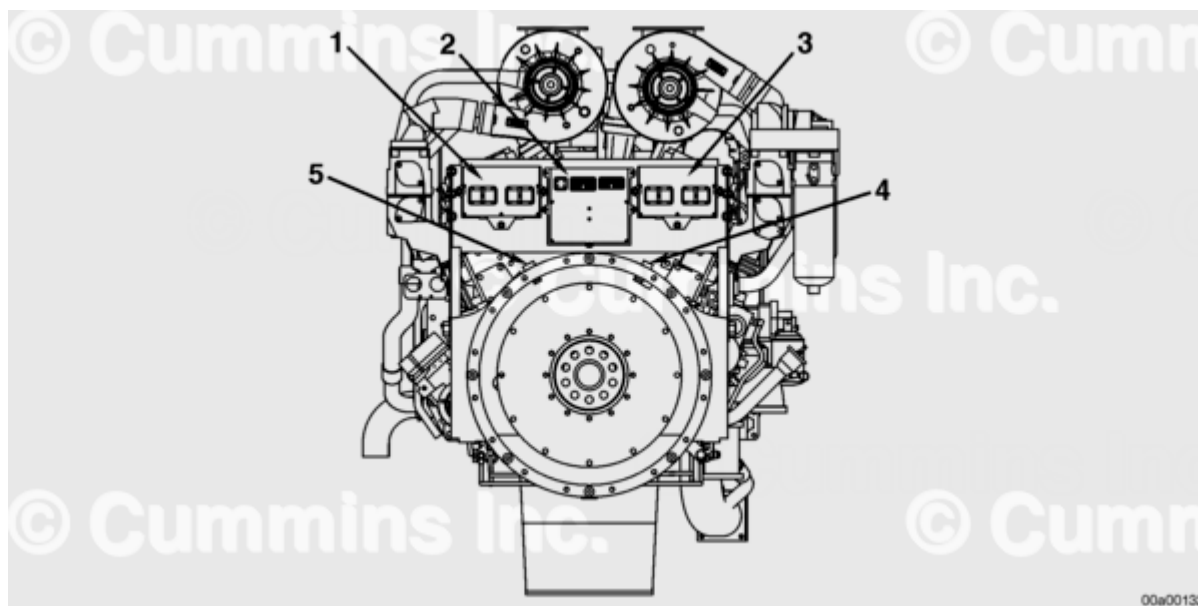
1. Alternator
2. Intake manifold 1 pressure sensor
3. Intake manifold 2 pressure sensor



Left View - QST30 PGI

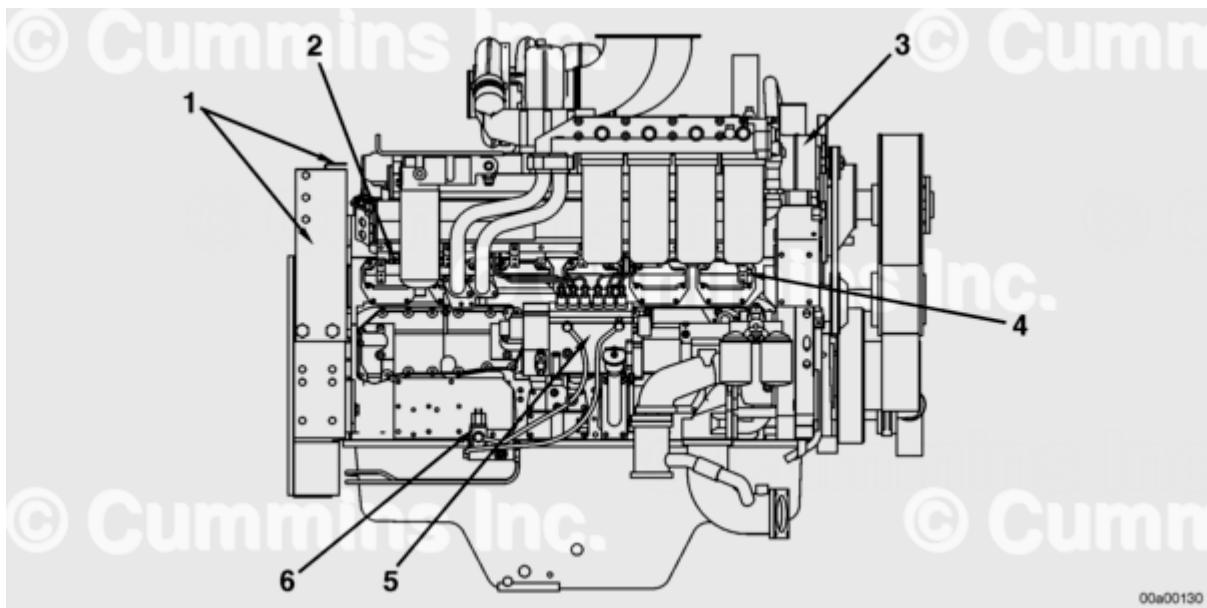
1. Coolant pressure sensor
2. Coolant temperature sensor
3. Barometric pressure sensor
4. Intake manifold 1 temperature sensor
5. ECM mounting bracket
6. Engine speed/position circuit

7. Engine oil temperature sensor 1
8. Oil rifle pressure 1 sensor
9. Starter
10. Engine fuel shutoff valve (left bank)
11. Fuel pump (left bank)
12. SAE J1939 data link service tool connector (CM850)
13. SAE J1939 data link service tool connector (CM552)
14. Intake manifold 1 pressure sensor



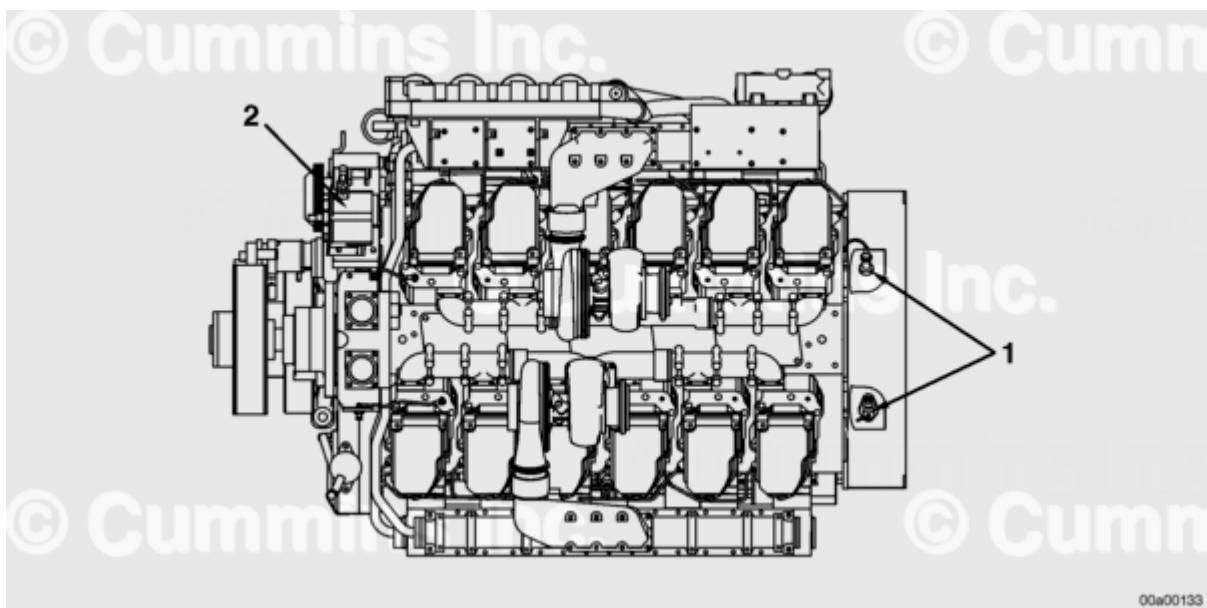
Rear View - QST30 PGI

1. ECM 1 (CM552)
2. ECM 2 (CM850)
3. ECM 3 (CM552)
4. Engine speed/position sensor
5. Engine speed/position sensor



Right View - QST30 PGI

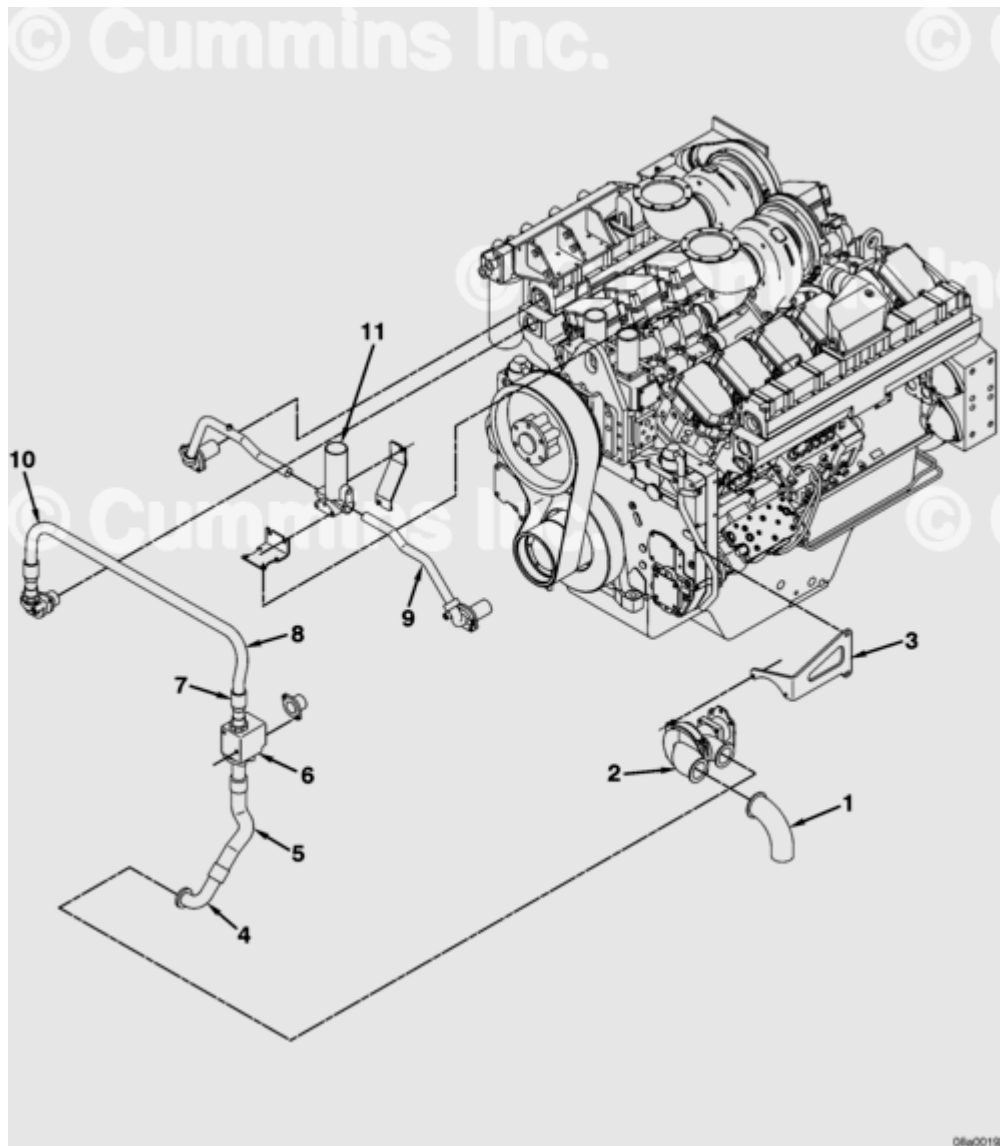
1. ECM mounting bracket
2. Intake manifold 2 pressure sensor
3. Alternator
4. Intake manifold 2 temperature sensor
5. Fuel pump (right bank)
6. Engine fuel shutoff valve (right bank)



Top View - QST30 PGI

1. Engine speed/position sensors
2. Alternator

Exploded View



Non-Thermostatically Controlled Two Pump Two Loop Cooling System Diagram

1. Low temperature aftercooled water pump inlet
2. Low temperature aftercooled water pump
3. Low temperature aftercooled water pump support bracket
4. Low temperature aftercooled water transfer hose
5. Low temperature aftercooled water transfer hose
6. Low temperature aftercooled water transfer connection
7. Low temperature aftercooled water transfer hose
8. Low temperature aftercooled water transfer hose
9. Low temperature aftercooled water transfer hose
10. Low temperature aftercooled water transfer hose
11. Low temperature aftercooled water outlet.

